14 Group decision making

Imagine you are a contestant on the popular TV game show ‘Who wants to be a millionaire?’ You have answered a few questions and have some money ‘in the bank’ but now you are facing a tricky question. You still have all three ‘lifelines’ in hand so to get help with the answer you can phone a friend, ask the studio audience or use the 50:50 to eliminate two of the four multiple-choice answers. Which lifeline should you opt for?

The choice between ‘phone a friend’ and ‘ask the audience’ requires deciding whether to rely on the intelligence of a single person or on the ‘wisdom of the crowd’ (Surowiecki, 2005). Intuitively, we might expect that the expert friend at home would be a better bet than the collection of individuals who just happen to be in the studio. Is this intuition correct? Surowiecki (2005) obtained statistics from the US version of ‘millionaire’ and discovered that in fact the opposite was true: experts were right on average 65 per cent of the time, but the audience picked the correct option on an impressive 91 per cent of occasions!

Surowiecki acknowledges that this anecdotal evidence would not necessarily stand up to scrutiny – for example it may simply be the case that the audience are asked easier questions than the experts – but the data do appear to suggest that several heads might be better than one when it comes to making certain types of decision. (An interesting footnote to Surowiecki’s observation is that the most successful contestant on the Australian version of the show (at the time of writing) used his ‘ask the audience’ lifeline on the penultimate and thus very difficult question, and chose the option voted for by the *fewest* members of the audience. Presumably, the contestant reasoned that for very difficult questions the obvious answer is often wrong so it is reasonable to go against the audience choice – he was right (or lucky) and went on to win the million dollars.) These observations from game shows are all very interesting, but as the saying goes, the plural of ‘anecdote’ is not ‘data’ – what do we know from controlled empirical tests about the merits or otherwise of group decision making?

Intellective and judgment tasks

The received wisdom concerning group decision making is that by working together on a problem we will arrive at a better solution than if we ponder the problem alone. Why else would we have invented juries, think-tanks or brainstorming sessions? However, the literature on group decision making does not always concur with this 'wisdom'. In a review of over 50 years' worth of research on group decision making, Hill (1982) concluded that group judgments were about as accurate as the second best individual member of the group. In a later analysis, Gigone and Hastie (1997) echoed the earlier conclusion, stating that: 'For the most part group judgments tend to be more accurate than the judgments of typical individuals, approximately equal in accuracy to the mean judgments of their members, and less accurate than the judgments of their most accurate member' (p. 153). The same basic conclusion was drawn by Kerr and Tindale (2004) in a recent review of the literature.

So what is the empirical basis for these conclusions about group performance? One class of problems commonly used to compare individual and group performance is known as 'eureka-type' problems or intellective tasks because they have a demonstrable solution (e.g., Laughlin, 1999; Lorge & Solomon, 1958; Maier & Solem, 1952). A good example of one of these problems is the rule induction task used by Laughlin and colleagues (Laughlin, 1999; Laughlin, Vanderstoep, & Hollingshead, 1991). Laughlin's task requires participants to induce a rule involving standard playing cards. The task begins with one rule-following card exposed face up on the table; participants are then asked to select a new card from the deck in order to test their hypotheses about the rule. For example, the eight of diamonds might be face up and the to-be-discovered rule might be 'Two diamonds followed by two clubs'. If a participant selects a card consistent with the rule, it is placed to the right of the first card; if it is inconsistent it is placed underneath the first card. Participants continue these trial-by-trial tests of their hypotheses and attempt to use the feedback to infer the rule. The interesting manipulation in the experiments reported by Laughlin et al. (1991) is whether participants are invited to test their hypotheses individually or as part of a cooperative four-person group.

Laughlin et al. (1991) found that the best individual participants generated significantly higher proportions of correct hypotheses than did the groups or second, third or fourth best individuals. The groups and second best individuals did not differ significantly from each other, but they did produce more correct hypotheses than the third and fourth best individuals (who did not differ significantly from each other). This pattern showing group performance to be equal to the second best individual is consistent with most previous research (e.g., Gigone & Hastie, 1997; Hill, 1982; Kerr & Tindale, 2004). Laughlin et al. conjectured that the poorer performance of the group may have been due to restrictions in the amount of evidence available for hypothesis testing and the time available to discuss potential rules. In a follow-up study they tested this idea by allowing groups 10 extra minutes to solve the problem and providing the opportunity to obtain more information about the rule on each trial. These manipulations led to equivalent performance for the groups and best individuals: both produced the same proportion of correct hypotheses. Thus it appears that given sufficient time and information groups can solve intellective tasks at least as well as the best of an equivalent number of individuals (Laughlin, 1999; Laughlin et al., 1991).

At the opposite end of the spectrum from intellective tasks are those commonly referred to as judgmental tasks. These tend to involve evaluative, behavioural or aesthetic judgments and have no demonstrable solution (e.g., sales forecasting) (Laughlin, 1999). How does the performance of groups and individuals compare on these types of task? Can the group perform as well as the best individual, as they seem to be able to do in the intellective tasks? A study reported by Sniezek (1989) addressed this question. Sniezek presented sales forecasting problems to four groups of five undergraduate students. The task involved predicting sales volumes for a general store on campus. The groups received time-series data for the preceding 14 months and were asked to predict sales for the following month. Sniezek was interested not only in the comparison of individual and group performance but also in different methods of group interaction.

First, all members of the group provided an independent individual sales estimate – these estimates were then collated to provide a 'collective mean' judgment for the group. Second, one of four different group decision techniques was imposed on the group: dictator, consensus, dialectic or Delphi. The dictator technique required group members to decide, through face-to-face discussion, who was the best member of the group and then to submit his or her estimate as the group estimate. The consensus technique was a straightforward discussion aimed at coming to group agreement on the estimate. For the dialectic technique members were provided with the collective mean estimate and then asked to think of all possible reasons why the actual sales volume might be higher or lower than the estimate, following this discussion a revised group estimate was decided on. Finally the Delphi technique required group members to provide estimates anonymously in a series of rounds, with no face-to-face discussion, until a consensus was reached. (This technique is supposed to maximize the benefits of group decision making and minimize possible adverse effects such as one person monopolizing discussion.)

To measure accuracy Sniezek looked at the absolute percent error (APE) between the collective mean estimate and the group estimates. All the group interaction techniques led to slightly more accurate forecasts than the simple aggregation of individual estimates. The greatest improvement was shown by the dictator group (reduction of 7.5 per cent) followed by the Delphi (2.3 per cent), dialectic (1.3 per cent) and consensus (0.8 per cent) groups. However, the APE reduction achieved by the best members was 11.6 per cent, indicating that the best members considerably outperformed all the group decision techniques. It is also worth noting that although groups seemed to be successful in identifying their best member – hence the relatively good performance of the dictator group – the dictators tended to change their judgment following group discussion, and these changes were all in the direction of the collective mean and hence more error. On average the final dictator estimates had 8.5 per cent higher APE than the initial ones.

Sniezek took care to point out that the generality of these results is not known. The groups were small, the participants were undergraduates, and the techniques were only tested in a single context (sales forecasting); nevertheless the results suggest that group interaction and discussion can sometimes lead to improvements in judgment accuracy – at least to a level that is better than the collective mean judgment. If this is the case, then it is important to consider how these interactions might occur – that is, how is the consensus achieved?

Achieving a consensus

Sniezek and Henry (1990) describe consensus achievement in terms of a revision and weighting model. They argue that this two stage model involves the conceptually distinct processes of revision and weighting, which can both operate to transform the distribution of individual judgments into a consensus group judgment. Sniezek and Henry suggest that the fundamental difference between these two processes is that revision occurs at the level of the individual within a group, whereas weighting (i.e., the combination of multiple judgments) occurs at group level. The evidence from two experiments that were similar in design to the Sniezek (1989) experiment discussed earlier, suggested that the weighting process was the more important one for achieving consensus and improving group accuracy (Sniezek & Henry, 1989, 1990). Social interaction of group members during the revision process had little appreciable impact on judgment accuracy; only when the revision process ended and the group engaged in weighting and combining multiple individual judgments were the improvements in accuracy observed.

Gigone and Hastie (1997) built on the ideas of the revision and weighting model, but introduced a Brunswikian lens model framework for conceptualizing the group judgment process. In chapter 3 we encountered the lens model and discussed how the idea of a 'lens of cues' through which a decision maker sees the world is a metaphor that has inspired many researchers (e.g., Hammond & Stewart, 2001). Gigone and Hastie extended the metaphor to think about how groups might arrive at consensus judgments. Figure 14.1 is a graphical representation of their model. Its similarity to the individual lens model shown in Figure 3.1 should be immediately apparent. The far left hand side in both diagrams represents the environment containing the to-be-judged criteria (C); the centre represents the cues that are probabilistically related to the criteria; the far right is the consensus group judgment (G) in Figure 14.1 and the individual judgment in Figure 3.1.

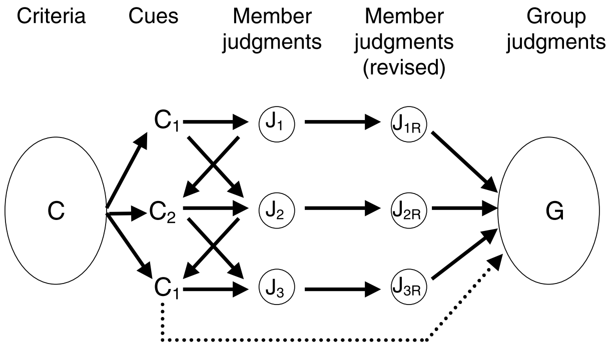

Figure 14.1 A lens model of the group judgment process. From Gigone, D., & Hastie, R. (1997). Proper analysis of the accuracy of group judgments. *Psychological Bulletin*, *121*, 149–167. Copyright 1997 by the American Psychological Association. Adapted with permission.

The clear difference between the individual model and the group model is that the latter has two extra layers or stages. In the group model the cues give rise to the initial member judgments and these are then revised before being combined into the group judgment. The advantage of using a lens model framework for thinking about group judgment is that the model lends itself readily to a variety of statistical techniques for analysing judgmental accuracy. For example, the 'lens model equation' (e.g., Cooksey, 1996), can be used to investigate the correlation between the best linear model of the environment (taking into account the validities of all the cues present) and the linear model used by each member of the group. Gigone and Hastie (1997) argue that a group's judgment accuracy depends fundamentally on the accuracy of individual member judgments, so examining individual judgments should be the starting point for understanding group performance.

The model can also be used to think about how accuracy is affected when the group convenes and discusses its judgment. For example, in group discussion a member might learn about a previously unknown judgment-relevant cue. If the member then weights this newly discovered cue appropriately the result will be an increase in the overall correlation between the member's judgment and the environment. Successful combination of members' judgments depends partly on the way in which errors are distributed across the group (Hogarth, 1978). If group judgments converge towards a member whose judgment is highly correlated with the environment then the group judgment will be accurate. However, if there is systematic bias in members' judgments, or a particularly persuasive individual in the group has low accuracy, then the combination process could result in poorer group judgment (Gigone & Hastie, 1997). Therefore if a group adopts an unequal weighting scheme (e.g., one that weights some members' judgments more highly than others) it is very important for the group to be able to identify which of its members are most accurate. Recall that the Delphi technique, in which there is no interaction and judgments are made anonymously, was developed, in part, to counteract the negative impact of domineering but potentially inaccurate group members.

One final aspect of the model in Figure 14.1 that warrants mentioning is the possible direct influence a cue can have on group judgments (depicted by the dashed line connecting the cues and the group judgment). There are at least two ways in which a cue might exert direct influence on the group judgment. One possibility is that there is an 'unshared' cue – that is, one that was only known by a single member of the group and did not come to light until the weighting and combination process. If that cue is a valid predictor then its addition in the group judgment policy will increase the overall correlation with the environment. A second possibility is that during discussion the group might decide a particular cue is very important and thus assign it more weight than did any group member in their individual judgments. Again, if the cue is valid correlational accuracy should improve.

These suggestions for how cue weights might be adjusted and integrated in group judgments are by no means exhaustive. As Plous (1993) notes it is somewhat ironic that the complexity and richness of group research can hinder theoretical progress. The variety of different variables (tasks, group sizes, group members, decision rules) used in experimental studies often makes it very difficult to compare results or draw general conclusions. Gigone and Hastie's (1997) model is very useful in this regard as it provides a framework for applying a common methodology and for improving the ability to make precise comparisons of individual and group accuracy.

Despite the difficulty in drawing general conclusions about group performance, one factor that does appear to be consistent in both intellective and judgmental tasks is the importance of identifying the best individual member of a group. In intellective tasks, like the rule-induction task described earlier, identification of the best member should be straightforward because the solutions to these tasks are demonstrable. Once one member has 'got it' (a 'eureka moment') he or she should be able to demonstrate the 'truth' of the solution. Indeed such a strategy is often described as a 'truth-wins' or 'truth-supported-wins' strategy (Hastie & Kameda, 2005). In judgmental tasks the identification of best members is more difficult because there is not a demonstrable solution. In such tasks groups have to rely on intuitions about a member's credibility or likelihood of being able to generate an accurate estimate (Henry, 1993). An everyday example of this kind of phenomenon can be seen in group attempts to come up with an answer in trivia quizes. Many of us will be familiar with the experience of being asked a general knowledge question at a trivia night and then trying to achieve a consensus by pooling the resources of the team. For example, if your team was asked 'How long is the river Nile?' different members might attempt to justify their own estimates by claiming that they have been to Egypt and seen the Nile, or that they know the Amazon is so many kilometers and that the Nile is longer, and so on. In an experimental study using general knowledge questions, Henry (1993) provided evidence that groups were able to identify their best members at levels far exceeding chance expectations, and that group members tended to engage in this process of identification as a normal part of the group judgment process.

One finding that appears to contradict the general pattern of accurate identification of best members comes from a study of group decision performance on a conjunction error problem. Recall from chapter 5 that the conjunction error is made when the probability of a conjunction, (e.g., *P*(heart attack and over 55), is rated as more probable than one of its conjuncts (e.g., *P*(heart attack)) (Tversky & Kahneman, 1983). Tindale, Sheffey, and Filkins (1990, as cited in Kerr, MacCoun, & Kramer, 1996) identified the number of persons in four member groups who did and did not commit a conjunction error in individual pretesting and then examined whether the group itself (following discussion of the problem) committed the error. Conjunction problems are intellective because they have a clear demonstrable solution – drawing a Venn diagram like the one shown in Figure 5.1 seems to convince most students – so one might expect that provided at least one member of each four-person group had not committed the error in pretesting, the group as a whole would not commit the error.

This was not what Tindale et al. (1990) found. Seventy-three per cent of the groups containing one member who had not committed the error in pretesting and three members who had, committed the error when making a group judgment. Even those groups with even numbers of members who had and had not committed the error as individuals fared poorly – 69 per cent still rated the conjunction as higher in probability.

Kerr et al. (1996) explained the results in terms of the normatively incorrect alternative (committing the error) exerting a 'strong functional pull' on groups. They argue that when a functional model of judgment (loosely defined as a 'conceptual system . . . that is widely shared and accepted in a population of judges' (p. 701)) operates in opposition to a normative model, the group discussion will tend to exacerbate any bias present in individual members. As we saw in chapter 5, conjunction problems are often interpreted in ways that are at odds with the normative model but consistent with everyday conceptions of language use (e.g., Gigerenzer, 1996; Hilton & Slugoski, 1986) and so it is quite plausible that a group member who knows the demonstrably correct answer might be persuaded that he or she has misinterpreted the question, thereby leading to a group tendency to commit the fallacy.

Groupthink: Model and evidence

One aspect of Tindale et al.'s (1990) results that was abundantly clear was that when all the members in a group committed the conjunction error in individual pretesting, the group had very little hope of coming up with the correct answer – 90 per cent of these groups committed the error. This tendency to be sure as a group that the best decision has been made, even when that decision is demonstrably incorrect, is one of the identifiable dangers of group decision making. Consensus seeking is a good thing, providing all relevant and valid evidence is considered in the discussion (e.g., the dialectic technique used by Sniezek, 1989), but when there is a tendency to seek evidence that serves only to confirm an initial hypothesis or to support a predetermined course of action, it can lead to disastrous consequences (see Wason, 1960).

The most provocative and influential work exploring such a tendency is that of Irving Janis. Janis (1972) coined the term 'groupthink' to describe the type of decision making that occurs in groups that are highly cohesive, insular and have directed leadership. Through the detailed analysis of a number of historical fiascos (the Bay of Pigs, President Johnson's decision to escalate the war in Vietnam, President Truman's decision to do the same in North Korea) and a comparison with major decisions that were successful (the implementation of the Marshall Plan after World War II), Janis identified the characteristics of groupthink-affected decision making: 'groupthink-dominated groups were characterized by strong pressures towards uniformity, which inclined their members to avoid raising controversial issues, questioning weak arguments, or calling a halt to soft-headed thinking' (Janis & Mann, 1977, p. 130).

More specifically, Janis (1972), proposed a model of groupthink with five antecedent conditions, eight symptoms of groupthink, and eight symptoms of defective decision making that result once groupthink has taken grip. The antecedents are those mentioned earlier: cohesiveness, insularity, directed leadership, along with two others – a lack of procedures for search and appraisal of information, and low confidence in the ability to find an alternative solution to the one favoured by the leader. The symptoms of groupthink are:

- the illusion of invulnerability creating excessive optimism and encouraging extreme risk taking;
- collective efforts to rationalize in order to discount warnings that might lead members to reconsider their assumptions;
- an unquestioned belief in the inherent morality of the group, inclining members to ignore the ethical or moral consequences of decisions;
- stereotyped views of rivals and enemies as too evil to warrant genuine attempts to negotiate;
- direct pressure on any members that express strong arguments against any of the group stereotypes;
- self-censorship of doubts or counterarguments that a member might have in order to create an illusion of unanimity within the group; and
- the emergence of self-appointed 'mindguards' who protect the group from adverse information that might shatter shared complacency about the effectiveness and morality of the group's decision.

The groupthink concept has had a huge impact in both the academic literature (see for example the 1998 special issue of the journal *Organizational Behavior and Human Decision Processes*, *73* (2)) and popular culture (a search via the internet search engine Google produced an incredible 1,520,000 hits for 'groupthink'; 1 September 2006). It is easy to understand the appeal of such a seductive concept. The eight symptoms of groupthink seem applicable to a vast variety of decision contexts, and appear to provide a useful framework for thinking about how defective decision making arises. Indeed, two of us were so struck by the similarities between the groupthink symptoms and the characteristics of the decision making leading up to the invasion of Iraq in 2003 that we wrote to the *Psychologist* magazine stating that, 'at the time of writing [March 2003] it may just be the heads of state and government who are the victims, but if war results many more heads may be lost to groupthink' (Newell & Lagnado, 2003, p. 176). Tragically our predictions were correct and post-invasion Iraq is now suffering from many of the symptoms of defective decision making that Janis identified as resulting from groupthink-dominated decisions (e.g., the failure to work out the risks of a preferred strategy and the failure to develop contingency plans). We were not alone in our assessment of the influence of groupthink on the decision to go to war. After the invasion the US Senate Intelligence Committee reported the following:

> Conclusion 3: The Intelligence Community suffered from a collective presumption that Iraq had an active and growing weapons of mass destruction (WMD) program. This 'group think' dynamic led Intelligence Community analysts, collectors and managers to both interpret ambiguous evidence as collectively indicative of a WMD program as well as ignore or minimize evidence that Iraq did not have active and expanding weapons of mass destruction programs. This presumption was so strong and formalized that Intelligence Community mechanisms established to challenge assumptions and group think were not utilized.
>
> (Source: http://intelligence.senate.gov/conclusions.pdf)

However, the seductiveness and applicability of the groupthink concept may also be its weakness. Aldag and Fuller (1993; Fuller & Aldag, 1998) have questioned whether groupthink is merely a convenient and overused label and argued that direct empirical support for the groupthink model is almost non-existent. In a characteristic quote they stated, 'Our contention is that even the most passionately presented and optimistically interpreted findings on groupthink suggest that the phenomenon is, at best, irrelevant. Artificially gathering a sampling of decision-relevant factors into a reified phenomenon has only resulted in the loss of valuable information' (Fuller & Aldag, 1998, p. 177). The lost information Fuller and Aldag refer to is the advances they suggest could have been made in understanding deficiencies in group decision making if so much research had not been constrained to fit within the groupthink framework. Fuller and Aldag go as far as suggesting that some researchers have unwittingly acted as virtual ‘mindguards’ of the groupthink phenomenon.

To explain the lasting popularity of groupthink Fuller and Aldag invoke a rather esoteric but nevertheless instructive metaphor. They describe groupthink as an ‘organizational Tonypandy’ after the Welsh mining village that was reportedly the site of violent riots in 1910–11. The official story of the riots describes hordes of miners, who were striking for better pay and conditions, clashing with police and soldiers in a series of bloody incidents. In fact there was apparently no serious bloodshed and some doubt that army troops were even involved. Despite these doubts, stories about the riots have been perpetuated over time, and in some circles have taken on legendary status. Fuller and Aldag (1998) argue that the same thing has happened with groupthink. Proponents of the phenomenon, despite being aware of the lack of empirical evidence supporting it, continue to ‘herald its horrors . . . building a mosaic of support from scattered wisps of ambiguous evidence’ (pp. 165–166).

The truth about groupthink probably lies somewhere in between the colourful language use by Fuller and Aldag (1998) and the rhetoric of its proponents. There is little doubt that the concept has proved to be very useful in focusing attention on the potential flaws of group decision making (e.g., Turner & Pratkanis, 1998), but the time to move on to testing other potential models of group functioning (e.g., the General Group Problem Solving model, Aldag & Fuller, 1993) and to break free from the constraints of groupthink theorizing is almost certainly long overdue.

Summary

Research on group decision making is both appealing and frustrating. The richness of the environments tends to make controlled empirical testing difficult and thus theory advances slowly. Most research is concentrated on eureka/intellective tasks and judgmental tasks. In both types of task, group performance tends to exceed the collective mean but not reach the level of the most talented individuals in the group. The Brunswikian lens model provides a useful metaphor for conceptualizing the processes of group discussion, consensus seeking and revisions of estimates in judgmental tasks. One of the dangers of blinkered consensus seeking is groupthink, a concept that has inspired a great deal of research despite claims that empirical evidence for the phenomenon is scarce.